DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

Table des Matières

Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a web application designed for football fans to discover, filter, and track upcoming matches. The system implements a data-driven pattern centered on real-time match updates and user-specific interest tracking, utilizing a relational mapping between users and match entities to persist following states across sessions.

1.2 Maturity Assessment

The project is currently in a state of REFINEMENT. While the functional requirements and entity relationships are well-defined, there is a high-severity structural gap regarding the absence of 'Goals & Objectives', leaving the primary business drivers undefined. Additionally, critical uncertainties remain regarding the authentication method and the specific data provider.

1.3 Technical Stack

- **External Data API**: Required for schedules, scores, and team information.
- **Database**: Relational storage for User and UserFollow entities.
- **Caching Layer**: Required to meet performance thresholds.

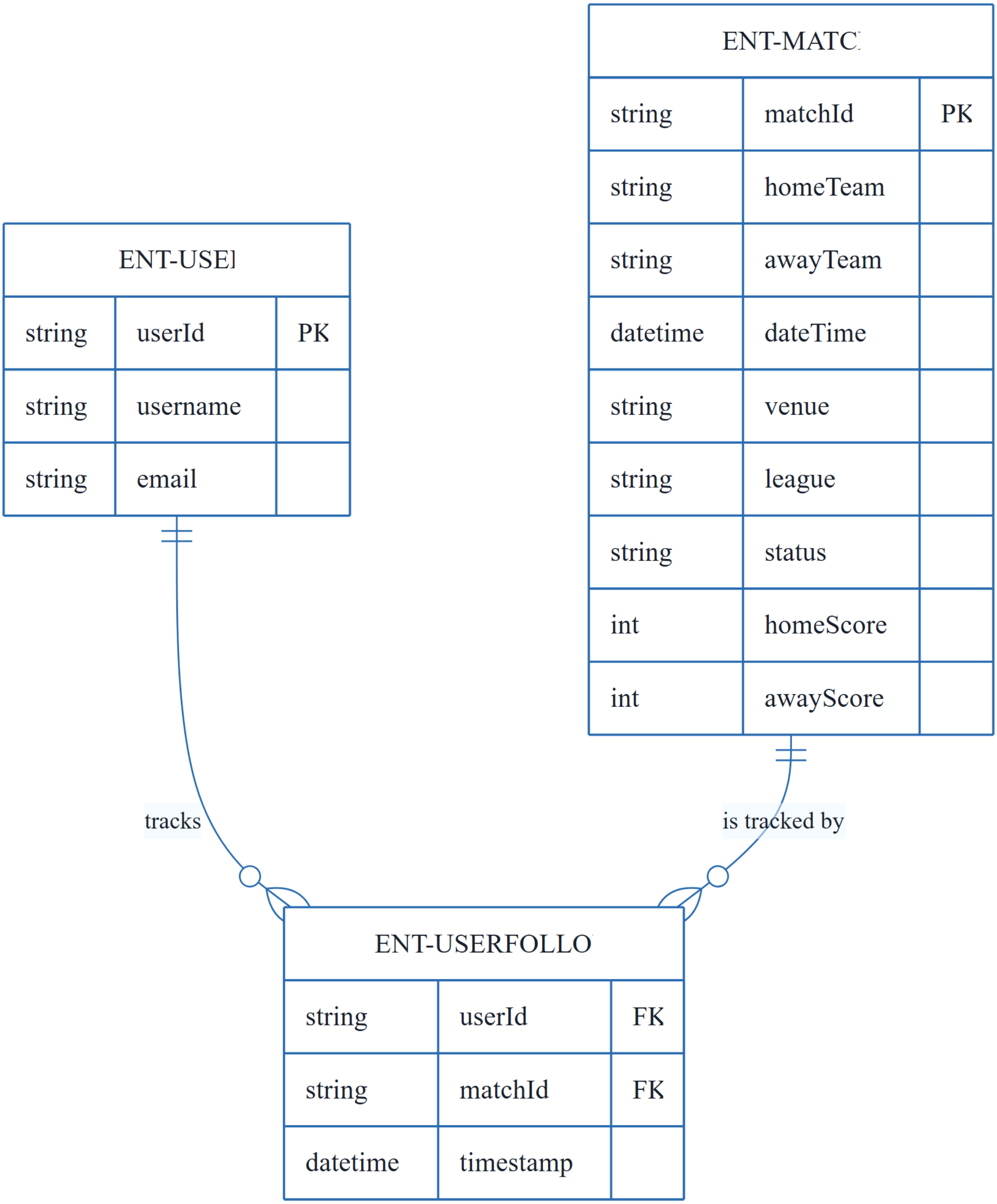
1.4 Architectural Constraints

- **Performance**: Match list load time must be < 2 seconds for 95% of requests using cached data.
- **MVP Scope Exclusion**: Social features, predictive analytics, and push notifications are strictly out of scope.

1.5 Critical Dependencies

- External Football Match Data API for schedules, scores, and team information.
- User authentication system for persisting UserFollow relationships.
- Referential integrity between UserFollow, User, and Match entities.
- Caching layer to meet the 2-second performance threshold.

2. Architecture Workflows


ENT-USE
string
userId
PK
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username
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email
ENT-MATC
string
matchId
PK
string
homeTeam
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awayTeam
datetime
dateTime
string
venue
string
league
string
status
int
homeScore
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ENT-USERFOLLO
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userId
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matchId
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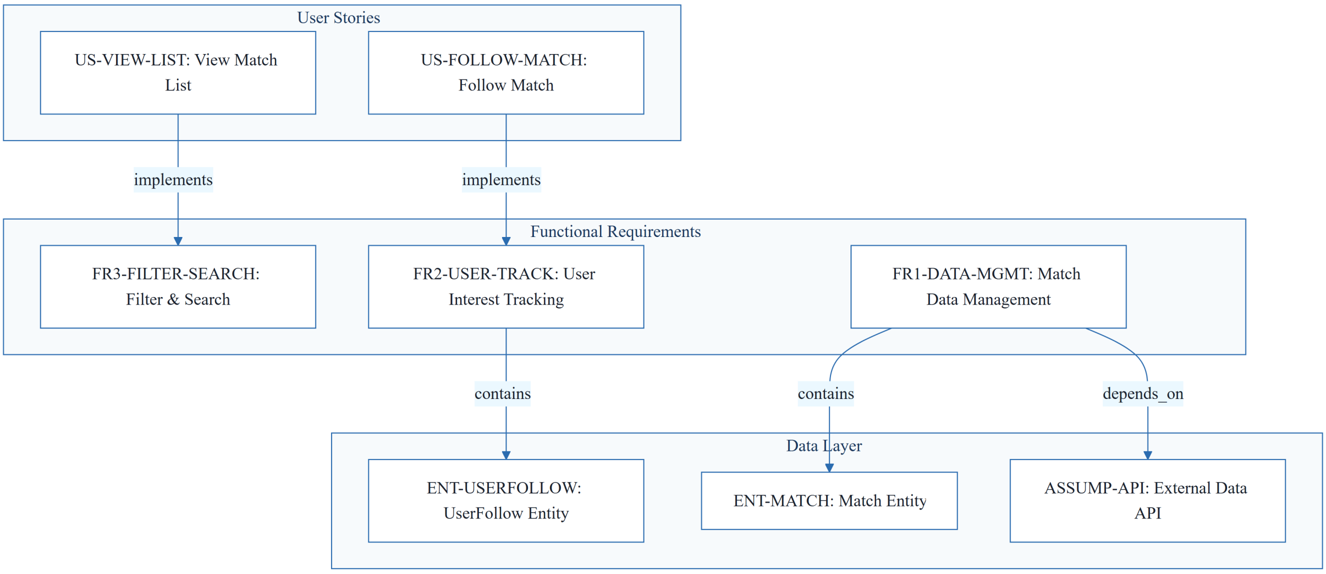
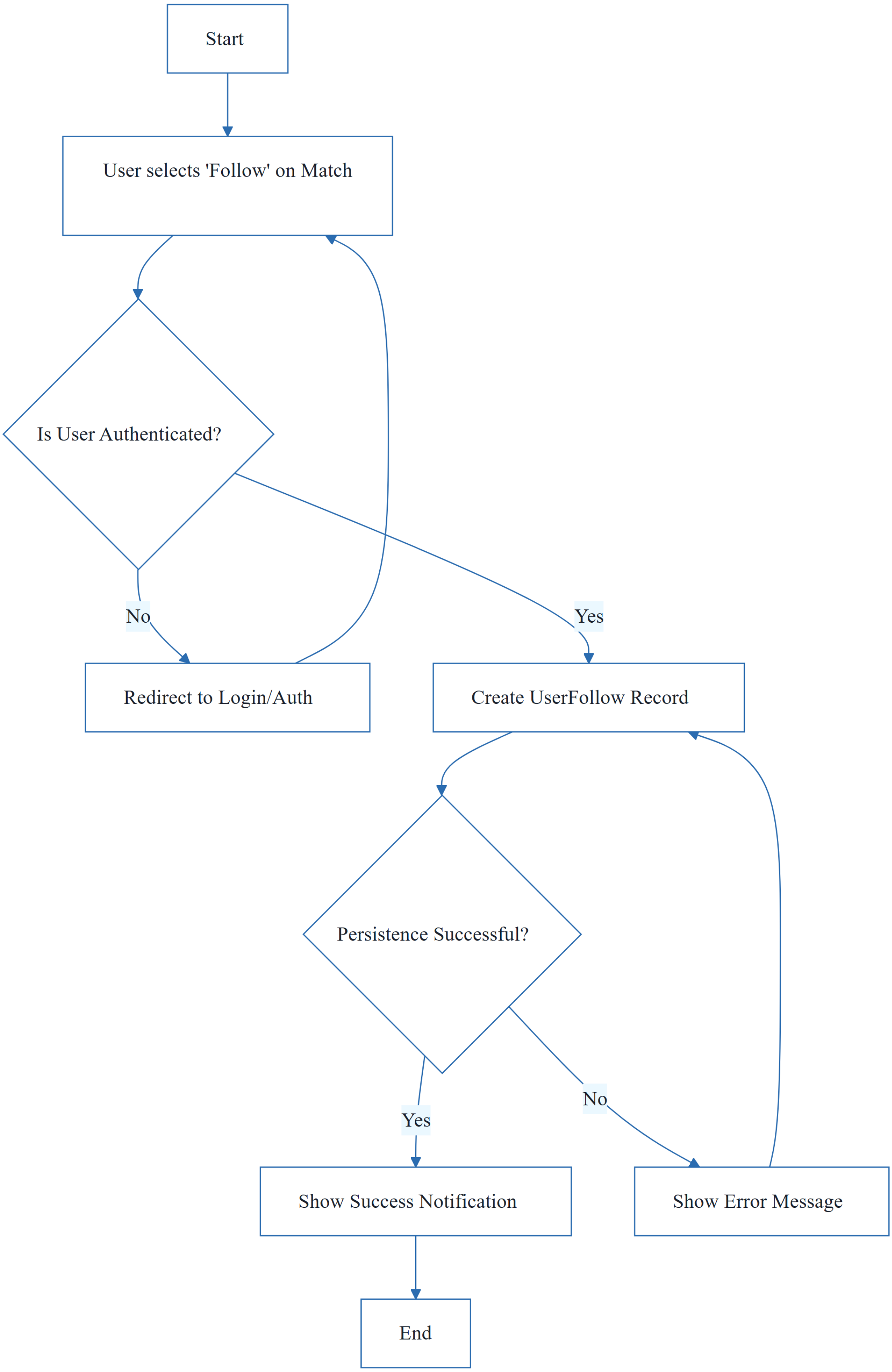
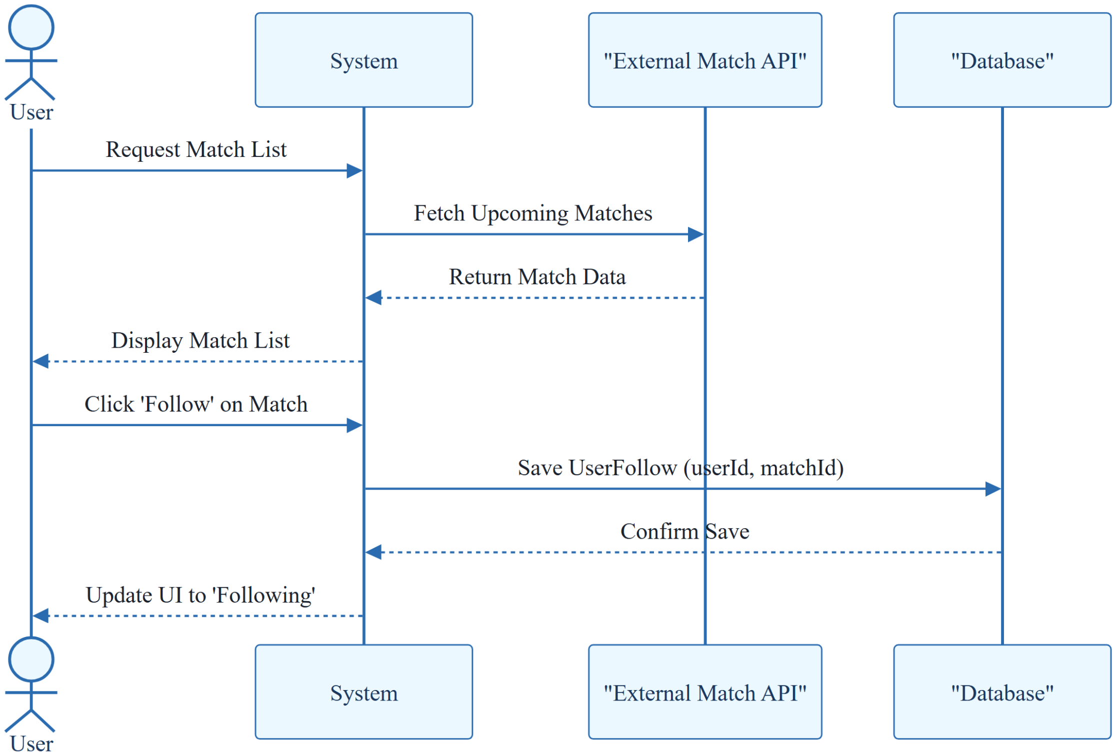
datetime
timestamp

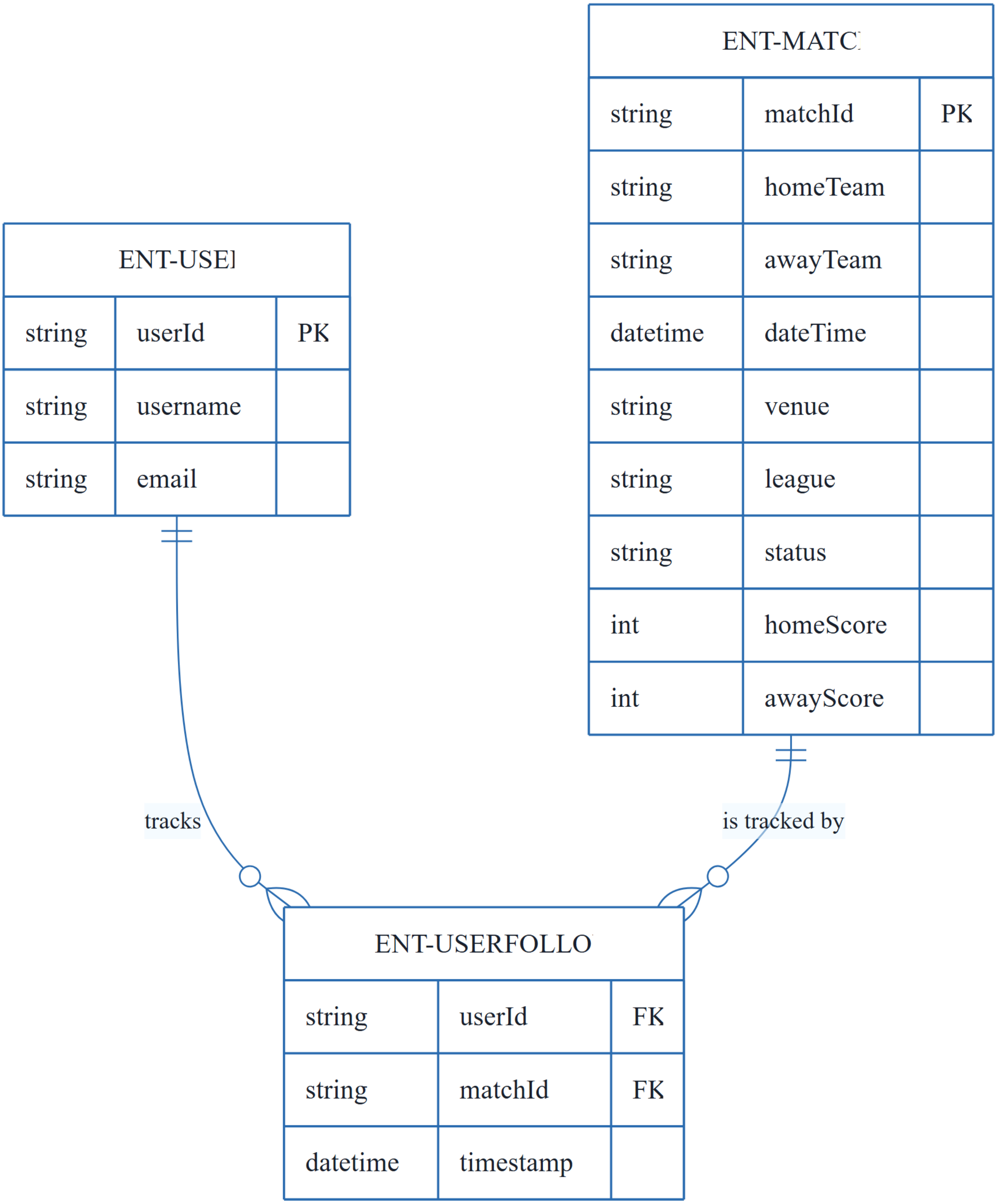

User Stories
US-VIEW-LIST: View Match List
US-FOLLOW-MATCH: Follow Match
implements
implements
Functional Requirements
FR3-FILTER-SEARCH: Filter & Search
FR2-USER-TRACK: User Interest Tracking
FR1-DATA-MGMT: Match Data Management
contains
contains
depends_on
Data Layer
ENT-USERFOLLOW: UserFollow Entity
ENT-MATCH: Match Entity
ASSUMP-API: External Data API


Start
User selects 'Follow' on Match
Is User Authenticated?
No
Yes
Redirect to Login/Auth
Create UserFollow Record
Persistence Successful?
Yes
No
Show Success Notification
Show Error Message
End

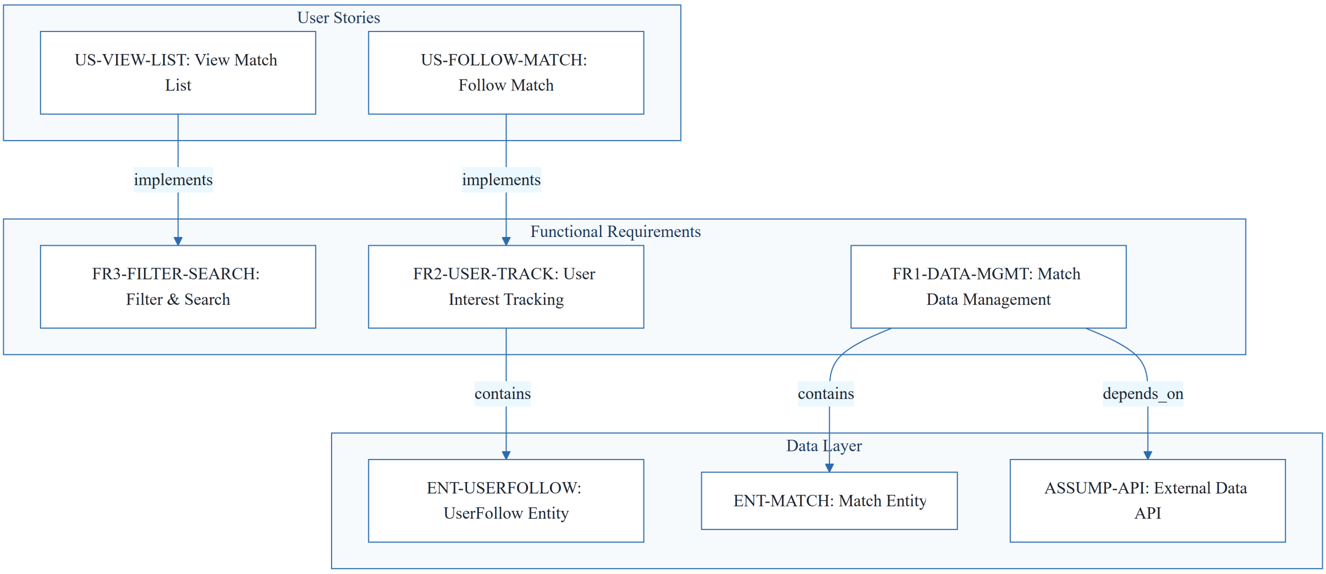


& Visual Diagrams

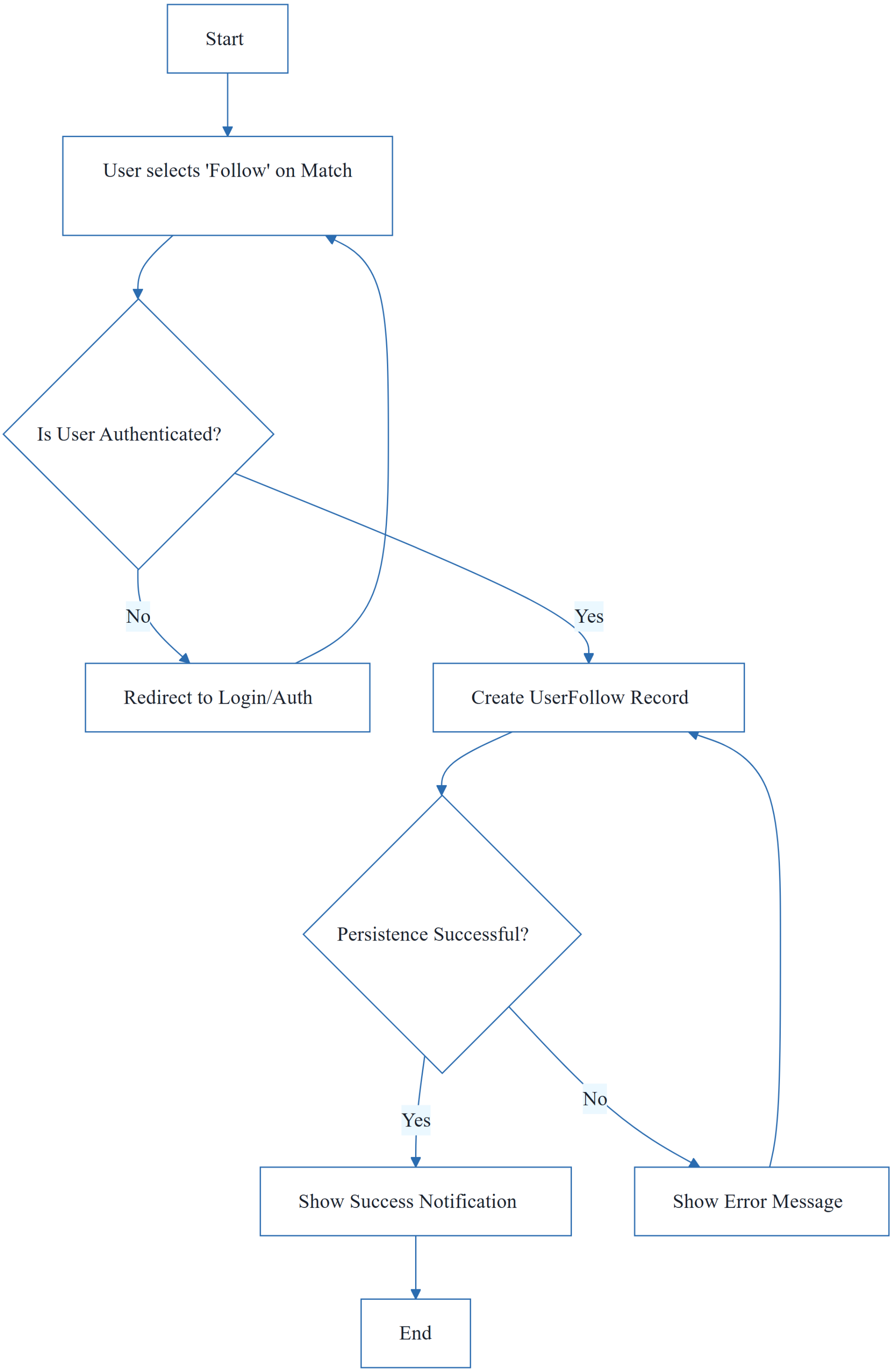
2.1 Football Match Manager Data Model



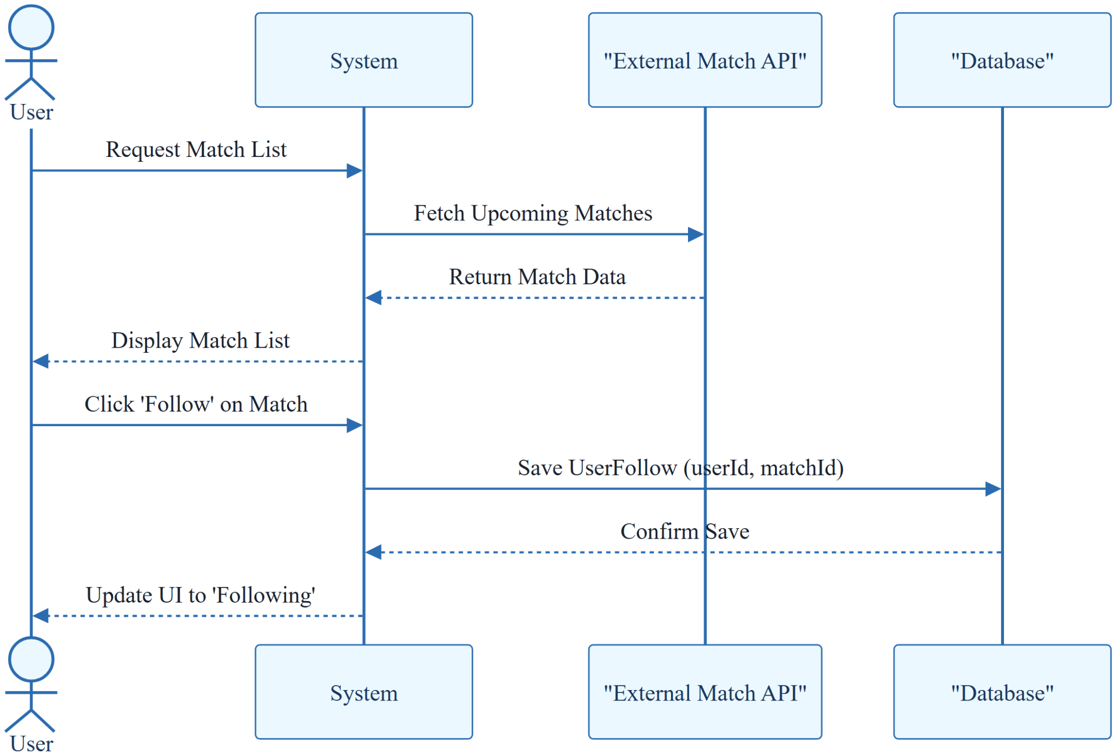
2.2 Requirements Traceability Matrix



2.3 Match Following Workflow



2.4 Match Tracking Interaction



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Relation / Dependency
US-VIEW-LIST	User Story	As a football fan, I want to see a list of upcoming football matches so that I can decide which matches to follow.	Implements FR3-FILTER-SEARCH
US-FOLLOW-MATCH	User Story	As a football fan, I want to mark a match as interesting so that I can easily track it later.	Implements FR2-USER-TRACK
FR1-DATA-MGMT	Functional Req	The system shall provide functionality to manage football match data including storage, retrieval via API, real-time updates, and archiving.	Contains ENT-MATCH; Depends on ASSUMP-API
FR2-USER-TRACK	Functional Req	The system shall allow authenticated users to follow/unfollow matches and persist these relationships across sessions.	Contains ENT-USERFOLLOW

ID	Type	Description	Relation / Dependency
FR3-FILTER-SEARCH	Functional Req	The system shall enable users to filter matches by league, date range, and search by team name.	-
NFR-PERF-LOAD	Non-Functional Req	Match list loads in under 2 seconds for 95% of requests (with cached data).	-
ENT-MATCH	Entity	Match: Attributes include matchId, homeTeam, awayTeam, dateTime, venue, league, status, homeScore, awayScore.	-
ENT-USER	Entity	User: Attributes include userId, username, email.	-
ENT-USERFOLLOW	Entity	UserFollow: Join entity linking userId to matchId with a timestamp.	Depends on ENT-USER, ENT-MATCH
ASSUMP-API	Assumption	Assume access to a reliable football match data API for schedules, scores, and team info.	-
CONS-MVP-SCOPE	Constraint	Social features, predictive analytics, and push notifications are out of scope for MVP.	-

3.2 Security Rules

- **Authentication**: Required for any action involving `ENT-USERFOLLOW` persistence.
- **Authorization**: Users may only modify their own `UserFollow` relationships.

3.3 Data Models

- **ENT-USER**: Primary identity record for application users.
- **ENT-MATCH**: Core data object retrieved from external API and cached locally.
- **ENT-USERFOLLOW**: Associative entity managing the M:N relationship between Users and Matches.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Goals & Objectives	HIGH	Add a section defining the primary business goal and the 'Why' behind the Football Match Manager to align stakeholders.

4.2 Remediation & Workflow

- **Immediate Action**: Define the high-level business objectives to resolve the structural gap.
- **Technical Decision Required**: Select the specific Football Data API provider.
- **Technical Decision Required**: Define the authentication method (e.g., OAuth2, JWT) for the MVP.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	ASSUMP-API	The external interface used for retrieving schedules, scores, and team information.
MVP	TECHNICAL_STACK	CONS-MVP-SCOPE	The initial version of the product excluding social features, predictive analytics, and push notifications.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	A join entity linking a specific person to a game with a timestamp to persist interest across sessions.